

# Ohio Public Health Information Warehouse

## Mortality Data Dictionary / Technical Notes

The Ohio Public Health Information Warehouse mortality data are available in reports and charts using two main case selection criteria:

1. **Resident Deaths** – Decedent was reported as being an Ohio resident, regardless of whether the decedent died in or outside Ohio.
2. **Occurrence Deaths** – Decedent was reported to have died within Ohio, regardless of whether the decedent was or was not an Ohio resident at the time of death.

This primary case selection criteria is built into the reports and can be deployed when choosing reports or charts (e.g., “Number of Resident Deaths by Year, Ohio”).

Depending on the specific report or chart selected by a user, a set or sets of additional analysis variables will be available. These choices are located in the upper left of the report builder interface. The variable choices are organized into categories based on the information contained in them. The categories and specific variable choices in the Warehouse mortality data are described in the following sections.

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## Demographics

**Age Group (19 Categories)** — Age in years at time of death is categorized in 19 age groups as follows: <1, 1-4, 5-9, 10-14, 15-24, ... 65-69, 70-74, 75-79, 80-84, 85+.

**Age Group (11 Categories)** — Age in years at time of death is categorized in 11 age groups as follows: <1, 1-4, 5-14, 15-24, 25-34, ... 65-74, 75-84, 85+.

**County of Residence** – The Ohio county where the decedent resided at the time of death (Adams, Allen, ... Wyandot).

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**Education Level** – Educational level attained by the decedent by the time of death: 8<sup>th</sup> grade or less; 9<sup>th</sup> through 12<sup>th</sup> grade (no diploma); High school graduate or GED; College, but no degree; Associate degree (e.g., AA, AS); Bachelor's degree (e.g., BA, AB, BS); Master's degree (e.g., MA, MS); Doctorate degree or professional degree; Unknown.

**Ethnicity** – The heritage, nationality group, lineage, or country of birth of a person or his parents or ancestors before their arrival in the United States. Decedents whose informants identify their origin as Spanish, Hispanic, or Latino may be of any race. Ethnicity is coded as Non-Hispanic, Hispanic (includes Mexican, Puerto Rican, Cuban, South or Central American, Dominican Republic, Spanish Not Otherwise Specified, Hispanic Not Otherwise Specified, Latino Not Otherwise Specified, and Other Specified Spanish/Hispanic (includes European)), and Unknown.

**Marital Status** – Marital status of decedent at time of death: Married (M), Married but separated (A), Widowed (W), Divorced (D), Never married (S), Not classifiable (U).

**Maternal Age Group** - Age in years at time of death is categorized in 9 age groups as follows: <15, 15-17, 18-19, 20-24, 25-29, ... 40-44, 45+.

**Neonatal Age Group** – An additional age grouping categorizes infant deaths into neonatal and post-neonatal ages: Neonates are decedents aged less than 28 days at time of death (N); Post-neonates are decedents aged 28-364 days at time of death (P), and all other deaths are not coded (U).

**Race** – The primary race of the decedent. An algorithm is used to assign multi-race decedent into a single race category for the Warehouse. The single race values are White, Black, American Indian or Alaska native, Asian or Pacific islander, Other, and Unknown.

**Sex** – Sex of decedent at time of death: Male, Female. When sex of the decedent is not known those deaths are not included in reports and the number of deaths omitted from the report due to missing data is shown in the footnote.

**Single Year Age** - Age in years at time of death is categorized by year as follows: 0, 1, 2, ... 82, 83, 84, 85+.

## Time/Location

**Disposition** – Disposition of the decedent's remains: Burial, Cremation, Donation, Entombment, Other, Removal from state, Unknown.

**Month of Death** – The month during which the decedent's death occurred.

**Place of Death** – The type of place where the death occurred: Hospital inpatient, Emergency room/outpatient, Dead on arrival at hospital, Decedent's home, Hospice facility, Nursing home/long term care facility, Other, Unknown.

**Resident State** – The decedent's state of residence: Ohio, Not Ohio.

**Occurrence State** – The state where the decedent's death occurred: Ohio, Not Ohio.

**Year** – The year during which the decedent's death occurred.

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### Cause of Death Variables

The Warehouse contains numerous variables derived from the text literal cause of death statements entered by the certifier for each decedent. There is space on the death certificate for the certifier to enter up to 20 distinct causes of death. Each cause is coded to an International Classification of Diseases, Revision 10 value (ICD code). A single primary underlying cause of death is determined for each death; this cause is the most important one for public health analysis. All the other causes that were entered for the decedent comprise the remainder of the multi-cause ICD 10 codes for the decedent. The primary underlying cause code and the multi-cause codes are categorized in many ways in the Warehouse. These cause grouping variable choices are described below:

### Causes of Death

**Detailed Causes of Death in 113 Groups** – This grouping of deaths into 113 categories is based on the primary underlying cause code and is used for most general mortality tabulations by cause. Additional technical details can be found in National Center for Health Statistics (NCHS) Instruction Manual 9 at this link: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

**General Causes of Death (39 Groups)** – This grouping of deaths into 39 categories is based on the primary underlying cause code and should be considered for analysis of smaller geographic areas. Additional technical details can be found in NCHS Instruction Manual 9 at this link: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

**Infant Causes of Death (130 Groups)** – This grouping of deaths into 130 categories is based on the primary underlying cause code and is developed for use in research about infant causes of death. This categorization scheme is customized to be most meaningful for use with infant deaths and provides much detail for those deaths. Users will need to filter on the age variable to limit the report to infants. Additional technical details can be found in NCHS Instruction Manual 9 at this link: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

**Leading Causes of Death (based on 113 list)** – This grouping of deaths into 50 categories is based on the primary underlying cause code and is the grouping that should be used when conducting leading causes of death analyses. Additional technical details can be found in NCHS Instruction Manual 9 at: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

**Leading Infant Causes of Death (based on 130 list)** - This grouping of deaths into 72 categories is based on the primary underlying cause code and is the grouping that should be used when conducting leading infant cause of death analyses. Users will need to filter on the age variable to limit the report to infants. Additional technical details can be found in NCHS Instruction Manual 9 at this link: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

**Manner of Death** – The fashion or circumstances that result in death: Natural, Unintentional injury (Accident), Suicide, Homicide, Could not be determined, Pending, Unknown. This variable is similar to the Intent of Injury in the Injury Indicators variable set. It differs in that the manner variable is assigned by the certifier and the Intent of Injury variable is based on the primary underlying cause code.

**Specific Causes of Death (358 Groups)** - This grouping of deaths into 358 categories is based on the primary underlying cause code and is the most detailed cause grouping provided in the Ohio Public Health Data Warehouse. Additional technical details can be found in NCHS Instruction Manual 9 at this link: [https://www.cdc.gov/nchs/nvss/instruction\\_manuals.htm](https://www.cdc.gov/nchs/nvss/instruction_manuals.htm)

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### COVID-19 Pandemic

The CDC recommends 'Coronavirus Disease 2019' or 'COVID-19' be reported on the death certificate for all decedents where the disease caused or is assumed to have caused or contributed to death. Additional CDC guidance is available at <https://www.cdc.gov/nchs/covid19/coding-and-reporting.htm>

The counts of COVID-19 deaths obtained in the Warehouse are based on cause of death statements contained on death certificates and will not exactly line up with other sources of COVID-19 mortality counts such as those based on COVID-19 testing. The Warehouse enables users to tabulate Ohio resident and occurrence deaths for custom requests. COVID-19 is coded to International Classification of Diseases, Revision 10, code U07.1.

### Injury Indicators

The injury (yes/no), intent of injury, mechanism (general) and mechanism (detailed) indicators are derived from the primary underlying cause codes. The traumatic brain injury variable is derived from the multiple cause codes. All injury indicators focus on the subset of deaths that are injuries. The intent of injury and the two mechanism variables are based on Centers for Disease Control Injury Matrix algorithms. More technical detail about the injury matrix is available at: [https://www.cdc.gov/nchs/injury/injury\\_matrices.htm](https://www.cdc.gov/nchs/injury/injury_matrices.htm)

**Injury Intent** – This variable is similar to the Manner of Death variable in the Causes of Death variable set. It differs in that the value in the intent of injury variable is based on the primary underlying cause code and not on the manner as assigned by the certifier (Coroner in most cases). The values in the intent of injury variable are Homicide, Suicide, Unintentional (injury), Natural (not an injury), Undetermined, and Legal Intervention or War.

**Mechanism (detailed)** – The Mechanism of External Injury (detailed) variable provides a grouping of injury deaths into the external mechanisms of injury (e.g., "firearm"). It has more granular detail concerning the mechanism than the Mechanism (general) variable. The categories in this variable cut across intent of injury types and enables users to study mechanisms of injury independent of intent of injury. The variable values are: cut or pierce, drowning, drug poisoning, fall, fire or flame, firearm, hot object or substance, machinery, MV-motorcyclist, MV-occupant, MV-other, MV-pedal cyclist, MV-pedestrian, MV-unspecified, natural, natural or environmental (injury), other land transport, other specified, other transport, overexertion, pedal cyclist-other (non MV), pedestrian-other (non MV), struck by or against, suffocation, unspecified.

**Mechanism (general)** – The Mechanism of External Injury (general) variable provides a grouping of injury deaths into the external mechanisms of injury (e.g., "firearm"). The categories in this variable cut across intent of injury types and enables users to study mechanisms of injury independent of intent of injury. The variable values are: cut or pierce, drowning, fall, fire or hot object or substance, firearm, machinery, motor vehicle traffic, natural, natural or environmental (injury), other land transport, other specified, other transport, overexertion, pedal cyclist-other (non MV), pedestrian-other (non MV), poisoning, struck by or against, suffocation, unspecified.

**Traumatic Brain Injury** – This variable is not part of the injury matrix methodology described above. The traumatic brain injury variable has the values of Yes and No. The value is yes if at least one of the decedent's multiple ICD 10 codes denotes a traumatic brain injury. (S01, S02.0, S02.2, S02.3, S02.7-S02.9, S04.0, S06-S07, S09.7-S09.9, T90.1, T90.2, T90.4, T90.5, T90.8, T90.9). Otherwise the value is No.

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### Disease Indicators

These variables were constructed to let users know when a specific chronic health condition is noted on the death certificate. These variables are yes/no indicators based on either the primary underlying cause or the multiple cause codes.

**Cancer Site/Type** - The Cancer Site/Type indicator variable is based strictly on the underlying cause of death, ICD code 113 categorization, where the category code has the following set of values: Bladder, Brain and Other CNS, Breast, Cervix, Colon and Rectum, Esophagus, Hodgkins Lymphoma, Kidney and Renal Pelvis, Larynx, Leukemia, Liver and Intrahepatic Bile Duct, Lung and Bronchus, Melanoma of Skin, Multiple Myeloma, Non-Cancer, Non-Hodgkins Lymphoma, Oral Cavity and Pharynx, Other Sites/Types, Ovary, Pancreas, Prostate, Stomach, Testis, Thyroid, Uterus.

**Chronic Disease** – The chronic disease indicator variable is based strictly on the underlying cause of death, ICD code 113 categorization, where the category code is in the following set of values:18-40, 43, 49-59, 61, 70-74, 83-86. It is No otherwise.

**Diabetes** – The value in this variable is Yes if diabetes is mentioned in the primary underlying cause on the death certificate (ICD codes E10 to E14 in primary underlying cause code). It is No otherwise.

**Enterocolitis C diff Indicator** – The value in this variable is Yes if enterocolitis due to *Clostridium difficile* is mentioned in the primary underlying cause on the death certificate (ICD code A04.7 in the primary underlying cause code). It is No otherwise.

### Poisoning Indicators

The poisoning indicator variables note when drug-specific overdoses and poisonings were mentioned anywhere on a death certificate. The indicator will be Yes when drug was mentioned and No otherwise. All but one of these variables are yes/no indicators based on the multiple cause codes and they are independently determined from the ICD codes for the death. One decedent may have multiple drug poisoning types mentioned. In those cases, each death will have a Yes in the poisoning indicator value for each drug type mentioned on the death certificate. The one variable that is not based on the ICD codes is the Fentanyl and Analogue indicator variable as there is not an ICD code that specifies fentanyl and its analogues. The Fentanyl and Analogue indicator is based on a text lookup for a positive mention of select text strings in any of the literal cause of death statements. This table shows which ICD 10 code was present to result in a Yes value in each poisoning indicator:

| Poisoning Indicator       | ICD 10 Code Present in the Multiple Cause Codes                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Alcohol (all types)       | T51.0-T51.9                                                                                                           |
| Alcohol (ethanol)         | T51.0                                                                                                                 |
| Alcohol (unspecified)     | T51.9                                                                                                                 |
| Barbiturates              | T42.3                                                                                                                 |
| Benzodiazepines           | T42.4                                                                                                                 |
| Cocaine                   | T40.5                                                                                                                 |
| Drug Induced              | T40.1-T41.1, T42.3, T42.4, T51.0, T51.1, T51.9                                                                        |
| Hallucinogens             | T40.7, T40.8, T40.9                                                                                                   |
| Heroin                    | T40.1                                                                                                                 |
| Methadone                 | T40.3                                                                                                                 |
| Methanol                  | T51.1                                                                                                                 |
| Opiates (all)             | T40.0-T40.4, T40.6                                                                                                    |
| Opiates (other)           | T40.2                                                                                                                 |
| Other Narcotics           | T40.6                                                                                                                 |
| Prescription Opiates      | T40.2-T40.4, T40.6                                                                                                    |
| Other Synthetic Narcotics | T40.4                                                                                                                 |
| Unspecified drugs         | T50.9                                                                                                                 |
| Fentanyl and Analogues    | Deaths related to fentanyl and its analogues have a positive mention of select text strings in the death certificate. |

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### Measures

The reports, charts, and maps present tabulations and calculations which are termed “measures”. Users must choose one or more measures to appear in reports.

**Age-Adjusted Rate** — A summary direct-adjusted rate that is a weighted average of the rates for each age group in the population (i.e. age-specific rates). The weights used in adjustment represent the age distribution of a standard population. Age-adjusted rates allow for [1] comparison of rates between populations with different age distributions and [2] comparison of rates within a population over time. The Ohio age-adjusted mortality rates are standardized to the age distribution of the U.S. 2000 standard population. This measure uses 11 age groups, calculates the age-specific rates for the age groups, then multiplies by the standard population proportion for the respective age group and sums to give an overall age-adjusted rate. Rates are presented in the Warehouse as the number of deaths per 100,000 persons.

**Age-Adjusted Rate (19)** — A summary direct-adjusted rate that is a weighted average of the rates for each age group in the population (i.e. age-specific rates). The weights used in adjustment represent the age distribution of a standard population. Age-adjusted rates allow for [1] comparison of rates between populations with different age distributions and [2] comparison of rates within a population over time. The Ohio age-adjusted mortality rates are standardized to the age distribution of the U.S. 2000 standard population. This measure uses 19 age groups, calculates the age-specific rates for the age groups, then multiplies by the standard population proportion for the respective age group and sums to give an overall age-adjusted rate. Rates are presented in the Warehouse as the number of deaths per 100,000 persons.

**Death Count** — The death count is the number of deaths. When a rate measure is requested the death count(s) are used as the numerator in the rate.

**Death Rate** — The death rate is the crude unadjusted mortality rate. The crude death rate is simply the number of deaths in a demographic group divided by the population in the demographic group times 100,000.

**Population** — The population is the number of residents for the demographic group that is being analyzed. When a rate measure is requested the population is used as the denominator in the rate.

#### ***Note on How to Obtain Age-Specific Rates:***

**Age-Specific Rate** — The number of deaths in a specified age group, per unit of population in that age group (i.e. per 100,000 persons), over a specified time period. Age specific rates can be produced in the Warehouse when the user uses age groups as a row selection and the death rate as the measure.